Job Description

Java Full Stack Engineer Trainee – Campus Hire

Role Title: Java Full Stack Engineer Trainee – Campus Hire

Program Context: Structured college-based training program aligned to Java Full Stack engineering readiness.

Role Summary

We are looking for high-potential campus hires from engineering institutions who will undergo a structured, college-based training program in Java Full Stack Development. The program is designed to build strong engineering fundamentals, hands-on coding capability, full-stack application development skills, and project readiness.

The trainee will be trained across backend, frontend, database, testing, DevOps awareness, secure coding, AI-augmented development, and Agile ways of working. By the end of the program, the trainee is expected to demonstrate the ability to design, develop, test, secure, and present a working full-stack application through a capstone project.

Key Learning Areas

1. Backend Engineering – Java & Spring Boot

Build strong backend development capability through Core Java, Modern Java, object-oriented programming, exception handling, collections, generics, memory management, Java 8–21 features, functional programming, streams, concurrency basics, and clean coding practices.

Develop capability in Spring Core, Spring Boot fundamentals, REST API development, validation, error handling, API documentation using OpenAPI/Swagger, Spring Data JPA, and Spring Security including JWT-based authentication and authorization.

2. Database & Data Handling

Learn relational database concepts, normalization, CRUD operations, SQL queries, joins, aggregations, views, indexing, constraints, transactions, ACID properties, and database interaction through Spring Data JPA.

3. Frontend Engineering

Gain practical exposure to HTML5, CSS3, Bootstrap, JavaScript, TypeScript, React.js, routing, state management, API integration, form handling, component lifecycle, hooks, frontend testing, and building responsive user interfaces.

4. Full-Stack Integration

Connect frontend and backend applications through REST APIs, manage authentication flows, handle validation errors across the stack, configure CORS, manage environment-based API URLs, and understand end-to-end data flow from UI to database and back.

5. Engineering Practices & Code Quality

Apply software engineering practices including Git-based version control, Maven build management, debugging, unit testing with JUnit, mocking with Mockito, integration testing, code coverage using JaCoCo, SonarQube awareness, logging, monitoring basics, SOLID principles, and commonly used design patterns.

6. DevOps, Cloud & Microservices Awareness

Receive foundational exposure to microservices concepts, inter-service communication, Docker, Docker Compose, CI/CD concepts, Jenkins pipeline awareness, Kubernetes concepts, and cloud fundamentals across major cloud service models and providers.

7. Secure Coding & Responsible AI Usage

Learn OWASP Top 10 vulnerabilities, secure coding practices, input validation, safe exception handling, protection of sensitive data, principle of least privilege, responsible AI, prompt engineering, GitHub Copilot usage, AI-assisted debugging, and critical review of AI-generated code.

Key Responsibilities During Training

- Participate actively in the structured Java Full Stack training program delivered in college.
- Build strong programming fundamentals using Core Java, Modern Java, OOP, collections, exception handling, and functional programming.
- Develop RESTful backend services using Spring Boot, Spring Data JPA, validation, exception handling, and security practices.
- Create responsive frontend applications using HTML, CSS, JavaScript, TypeScript, React.js, routing, forms, and API integration.
- Write clean, readable, maintainable, and testable code following coding standards and best practices.
- Use Git for version control and collaborate through branches, commits, pull requests, and code reviews.
- Perform unit testing, integration testing, and basic frontend testing using appropriate tools and frameworks.
- Apply debugging techniques using IDE tools, breakpoints, logs, stack traces, and AI-assisted troubleshooting.
- Demonstrate awareness of microservices, Docker, CI/CD, cloud fundamentals, logging, monitoring, and secure coding.
- Complete all checkpoints, assessments, coding exercises, and the final full-stack capstone project.
- Present the capstone project through demo, code walkthrough, Q&A, and peer review.

Required Technical Skills / Learning Outcomes

By the end of the training, the campus hire should be able to:

- Write Java programs using OOP, collections, exception handling, generics, streams, lambdas, and modern Java features.
- Design and build REST APIs using Spring Boot.
- Implement database operations using SQL and Spring Data JPA.
- Apply basic authentication and authorization using Spring Security and JWT.
- Build frontend applications using React.js, TypeScript, JavaScript, HTML, CSS, and Bootstrap.
- Integrate frontend applications with backend REST services.
- Write unit, integration, and component-level tests.

- Follow clean code, SOLID principles, design patterns, logging, and coding standards.
- Understand secure coding practices and common vulnerabilities.
- Use Git, Maven, IDE debugging, Docker basics, and CI/CD concepts.
- Use AI tools such as GitHub Copilot responsibly for code generation, debugging, test creation, and productivity enhancement while validating correctness and quality.

Behavioural & Professional Skills

In addition to technical capability, the trainee is expected to demonstrate the following behavioural competencies:

1. Articulation & Communication

- Clearly explain technical concepts, design decisions, code logic, and implementation approach.
- Provide structured status updates on assigned tasks, progress, risks, and blockers.
- Write clear pull request descriptions, project notes, and technical documentation.
- Present project outcomes confidently during demos, walkthroughs, and panel reviews.

2. Learnability & Growth Mindset

- Show curiosity and willingness to learn new technologies, tools, frameworks, and engineering practices.
- Actively seek feedback and demonstrate improvement across assignments and assessments.
- Adapt quickly to evolving training requirements and project expectations.

3. Problem Solving

- Break down technical problems logically and approach them with structured thinking.
- Debug issues independently using logs, stack traces, test results, and available tools.
- Evaluate multiple solution options and choose simple, maintainable approaches.

4. Collaboration & Team Orientation

- Work effectively with peers during coding exercises, reviews, and capstone project activities.
- Participate constructively in discussions, peer reviews, and team-based problem solving.
- Respect coding standards, review comments, and shared engineering practices.

5. Ownership & Accountability

- Complete assigned learning modules, coding tasks, checkpoints, and assessments within expected timelines.
- Take ownership of code quality, testing, documentation, and final deliverables.
- Raise blockers early and follow through on commitments.

6. Professional Discipline

- Demonstrate punctuality, consistency, attention to detail, and adherence to training expectations.
- Follow ethical coding practices, responsible AI usage, and secure handling of data.
- Maintain a professional approach in communication, collaboration, and review discussions.